

MSDS 7330

File Organization and Database Management

Midterm Exam

Question 1: Develop and Manage a Basketball Tournament Database (Total: 100 points)

The association hosts multiple basketball tournaments annually in different locations. Teams can participate in multiple tournaments, and each tournament consists of multiple games. Below is the **minimum set of attributes** required for the database, but you are encouraged to **add relevant keys and constraints** to ensure proper normalization.

Required Attributes:

- **Tournaments:** TourneyID, TourneyDate, TourneyLocation
- **Teams:** TeamID, TeamName, CoachID
- **Coaches:** CoachID, CoachFirstName, CoachLastName
- **Players:** PlayerID, FirstName, LastName, Address, City, State, ZipCode, TeamID
- **Games:** GameID, TourneyID, HomeTeamID, AwayTeamID, WinningTeamID, CourtID
- **Scores:** GameID, PlayerID, PlayerScore
- **Courts:** CourtID, CourtLocation

Tasks to Complete:

1.1 Database Schema Normalization (10 pts)

- Develop a **fully normalized schema** (at least **3rd Normal Form (3NF)**).
- Ensure relationships are properly structured using **Primary Keys and Foreign Keys**.
- Create and provide an **ER Diagram** using **MySQL Workbench**. (Please do not draw it by hand)

1.2 Create the Database Using Forward Engineering with MySQL Workbench (10 pts)

- This will generate SQL DDL statements for **table creation**, create screenshot for submission

1.3 Insert Data into the Database from Provided CSV Files (10 pts)

- Load the data from the provided **CSV files** into your **MySQL database**.
- Submit SQL **INSERT statements** along with a sample query output.

1.4 Write SQL Queries as Stored Procedures and Call Them from Python (70 pts)

For each query, create a **stored procedure in MySQL** and **call it using Python**.
Submit:

1. **SQL Stored Procedure Code**
2. **Python Code Calling the Procedure**
3. **Screenshot of Results**

Queries:

1. **Retrieve locations** where tournaments are hosted.
2. **Display all players** with their **addresses formatted for mailing**, sorted by **zip code**.
3. **List teams and their coaches**.
4. **Show tournaments that have not been played yet**.
5. **Display the top 10 players** with the highest **total scores**.
6. **For each player, display their highest single-game score**.
7. **Find players whose highest score in a game is more than 10 points higher than their average score**.
 - *Hint: Use `MAX(PlayerScore)`, `AVG(PlayerScore)`, and a subquery/self-join.*

1.5 Create a View for Tournaments Held at 'Red Rooster Arena' (3 pts)

- The view should include:
 - `TourneyID`, `TourneyDate`, `TourneyLocation`, and total games played.
 - Filter only tournaments where at least **one game** was played at `Red Rooster Arena`.

Data Provided:

- **CSV files** containing data for **Tournaments, Teams, Players, Scores, and Games**.
- The **column names have been standardized** for consistency with SQL requirements.

What to Submit:

ER Diagram: Screenshot of **database schema**.

SQL DDL: SQL **table creation statements**.

Data Insertion: SQL **INSERT** statements.

Stored Procedures:

- **SQL Code** for each stored procedure.
- **Python Code** calling the procedures.
- **Screenshots of query results**.
- **View Creation:** SQL code for **Red Rooster Arena** view.

Submission Guidelines:

Submission Method: Submit all required files via **Canvas** (NO email submissions).

Best of luck! If you have any questions, reach out before the deadline.